

Session Plan

		Programme Batch Division Semester Course Code Course No of Credits Course Type Facilitator	UG 2025-29 A B C D E and F 2 TE7689 Statistics and Probability 3 Statistics and Probability Dr. Gopal Kumar Gupta													
CO No		CO Description									CO Performance Standard (%)					
Course Outcome (CO) 1		Understand the statistical data and their graphical representation wherever relevant.									60.00%					
Course Outcome (CO) 2		Interpret the data using appropriate measures of central tendency.									60.00%					
Course Outcome (CO) 3		Find measures of dispersion and compare two series using coefficients of variation									50.00%					
Course Outcome (CO) 4		Find coefficient of correlation and rank correlation									40.00%					
Course Outcome (CO) 5		Explain the line of best fit and estimate the value of dependent variable using regression analysis									60.00%					
Course Outcome (CO) 6		Define the concept of random variables and find the probability distribution									60.00%					
Recommended Books		Gupta nad Kapoor - Fundamental of Statistics														
Journals																
Other Support Material / Sources																
Session No	Topic	Subtopic	Duration (Hours)	Session Objective (SO)	Pre-Class Home work	Nature of Session/ Lecture/ Tutorial/ Project)	Session Methodology	Post Class reflective Work	Assessment Task No	Contact Hour Equivalent	CO1 Mapping with SO	CO2 Mapping with SO	CO3 Mapping with SO	CO4 Mapping with SO	CO5 Mapping with SO	CO6 Mapping with SO
1	Statistics Mathematical	Importance, scope and limitations of statistics	1	To introduce the concept of statistics and explain its significance, applications, and limitations.	Read introductory chapter on statistics from textbook	Lecture	Explanation with examples and discussion	Write summary of importance and scope of statistics	1	1	Strong Linkage	No Linkage	No Linkage	No Linkage	No Linkage	No Linkage
2	Statistics Mathematical	Primary and secondary data, Attributes and variables, data.	1	To distinguish between primary and secondary data and explain the nature of variables and attributes.	Watch video on types of data collection	Lecture	Interactive session using real-life examples	Identify examples of primary and secondary data around you	2	1	Strong Linkage	No Linkage	No Linkage	No Linkage	No Linkage	No Linkage
3	Statistics Mathematical	Examples	1	To apply the learned concepts through examples of data classification and representation.	Review class notes from previous session	Tutorial	Group activity solving practice problems	Solve two additional textbook problems	3	1	Strong Linkage	No Linkage	No Linkage	No Linkage	No Linkage	No Linkage
4	Statistics Mathematical	Examples	1	To reinforce understanding through additional practical examples of data types and collection.	Revise concepts of variables and data	Tutorial	Practice worksheet and peer discussion	Submit written solution of given worksheet	4	1	Strong Linkage	No Linkage	No Linkage	No Linkage	No Linkage	No Linkage
5	Statistics Mathematical	Discrete and continuous variables Definition of grouped	1	To explain the classification of variables and demonstrate how data is grouped and represented.	Read textbook sections on variables and classifications	Lecture	Chalk & talk with visuals	Write a paragraph on differences between discrete/continuous	5	1	Strong Linkage	No Linkage	No Linkage	No Linkage	No Linkage	No Linkage
6	Statistics Mathematical	Graphical Representations or Histogram, Ogive curves, stem and leaf charts.	1	To develop skills for visual representation of data using different graphical methods.	Read examples of graph types and their uses	Lecture	Use of charts, demonstration on Excel	Practice drawing graphs for a sample dataset	6	1	Strong Linkage	No Linkage	No Linkage	No Linkage	No Linkage	No Linkage
7	Statistics Mathematical	Examples	1	To practice drawing and interpreting data through graphical tools using real datasets.	Solve at least two problems using histogram and ogive	Tutorial	Solving graphical problems in class	Compare different types of graphical representation	7	1	Strong Linkage	No Linkage	No Linkage	No Linkage	No Linkage	No Linkage
8	Statistics Mathematical	Examples	1	To provide further hands-on practice with visual representation of various data sets.	Review notes on data types and graphs	Tutorial	Problem solving with peer discussion	Complete practice problems and submit answers	8	1	Strong Linkage	No Linkage	No Linkage	No Linkage	No Linkage	No Linkage
9	Statistics Mathematical	Arithmetic mean, median and mode, Properties and relations of mean, median and mode.	1	To introduce measures of central tendency and explain their properties and interrelationships.	Read chapter on central tendency measures	Lecture	Expansion with formulas and real-life examples	Solve 3 problems each on mean, median, and mode	9	1	Strong Linkage	No Linkage	No Linkage	No Linkage	No Linkage	No Linkage
10	Statistics Mathematical	Range, Quartile, Quartile deviation, Percentile, Mean deviation, Standard deviation, Variance.	1	To explain and compare various measures of dispersion and variability used in statistical analysis.	Read definitions and formulas; bring calculator	Lecture	Demonstration with dataset + problem solving	Attempt 5 problems using given formulas	10	1	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage	No Linkage
11	Statistics Mathematical	Examples	1	To apply the concepts of central tendency and dispersion to solve practical problems using real or hypothetical data.	Solve basic problems on variance and quartiles	Tutorial	Group practice on mixed problem sets	Submit solved worksheet with correct formatting	11	1	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage	No Linkage
12	Statistics Mathematical	Moments, Skewness and kurtosis.	1	To understand the significance of skewness and kurtosis in describing data distribution.	Watch video tutorial on skewness concepts	Lecture	Board explanation with graph illustrations	Draw skewness graph and compute using given data	12	1	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage	No Linkage
13	Statistics Mathematical	Examples	1	To reinforce understanding of distribution shape using practical examples and calculations.	Attempt numerical problems from previous session	Tutorial	Practice and peer evaluation of solutions	Peer-review at least two classmates' solutions	13	1	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage	No Linkage
14	Statistics Mathematical	Examples	1	To provide additional problem-solving practice with measures of shape and variability.	Revise moments and kurtosis formulas	Tutorial	Worksheet practice with guidance	Complete 3 unsolved questions from textbook	14	1	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage	No Linkage
15	Statistics Mathematical	Merits and demerits of statistical measures.	1	To critically evaluate the advantages and limitations of statistical measures in different contexts.	Read theoretical pros/cons of statistical tools	Lecture	Discussion + comparative analysis chart	Write your opinion on suitable measures for different datasets	15	1	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage	No Linkage
16	Statistics Mathematical	Examples	1	To provide additional practice in applying statistical measures through worked-out problems and exercises.	Revise formulas from previous session	Tutorial	Worksheet solving and discussion	Submit completed worksheet	16	1	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage	No Linkage
17	Correlation and regression	Correlation: introduction or coefficient, Karl Pearson's coefficient of correlation, short-cut method, Step deviation method for coefficient of	1	To introduce the concept of correlation and calculate Karl Pearson's coefficient to measure linear relationship.	Read intro to correlation, note real-life uses	Lecture	sign-negatives explanation with examples	Write interpretation of correlation value from example	17	1	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage	No Linkage
18	Correlation and regression	Examples	1	To explain and apply simplified methods for computing correlation efficiently.	Watch method tutorials, bring calculator	Lecture	Step-by-step method demo + blackboard solving	Solve 2 problems using each method	18	1	No Linkage	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage
19	Correlation and regression	Examples	1	To develop proficiency in correlation analysis using various computational techniques.	Solve Pearson coefficient problems	Tutorial	Peer discussion and checking	Submit error analysis on incorrect attempts	19	1	No Linkage	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage
20	Correlation and regression	Rank correlation	1	To understand and apply Spearman's rank correlation coefficient for ordinal data.	Read Spearman's formula, learn ranks	Lecture	Real data ranking and coefficient calculation	Apply method to class performance ranks	20	1	No Linkage	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage
21	Correlation and regression	Regression: introduction or coefficient, Angle of regression, Lines of regression.	1	To introduce the concept of regression analysis and derive regression lines.	Review regression meaning and scatter plots	Lecture	Graphical and analytical explanation	Solve 1 example using both X on Y and Y on X	21	1	No Linkage	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage
22	Correlation and regression	Examples	1	To reinforce understanding of regression lines through practical examples.	Try solving basic regression lines	Tutorial	Practice with datasets	Complete two new questions from assignment sheet	22	1	No Linkage	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage
23	Correlation and regression	Examples	1	To provide further practice with regression problems to build analytical skills.	Practice previous regression problems	Tutorial	Solve word problems using regression	Share summary of method in your own words	23	1	No Linkage	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage
24	Correlation and regression	Properties of regression coefficient, Angle between two regression lines.	1	To explore the mathematical properties of regression coefficients and understand their geometrical interpretation.	Read textbook section on regression properties	Lecture	Explain using angle formulas and diagrams	Solve problems involving regression coefficient relationships	24	1	No Linkage	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage
25	Correlation and regression	Examples	1	To deepen understanding of regression analysis through theoretical and graphical insights.	Revise last class examples and notes	Lecture	Derivation and numerical examples	Write summary of geometrical interpretation	25	1	No Linkage	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage
26	Correlation and regression	Examples	1	To apply the theoretical concepts of regression in solving numerical problems.	Solve regression coefficient problems	Tutorial	Practice from question bank	Submit accuracy-checked worksheet	26	1	No Linkage	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage
27	Correlation and regression	Examples	1	To practice and reinforce regression computations and interpretation.	Review scatter plot-based regression	Tutorial	Worksheet and board-based corrections	Reflect on difficulties in solving and share strategies	27	1	No Linkage	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage
28	Correlation and regression	Introduction to probability.	1	To introduce fundamental concepts of probability and its role in statistical inference.	Read probability basics and types of events	Lecture	Coin, die, card-based example introduction	Write down sample space examples from real life	28	1	No Linkage	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage
29	Correlation and regression	Examples	1	To apply basic probability rules to solve simple events and compound events problems.	Solve basic event probability problems	Tutorial	Group activity: scenario-based problem solving	Document group results and share insights	29	1	No Linkage	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage
30	Correlation and regression	Examples	1	To reinforce probability concepts through real-life and mathematical examples.	Try questions on simple & compound events	Tutorial	Practice quiz and revision	Evaluate self-performance and identify weak areas	30	1	No Linkage	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage
31	Probability Distribution	Conditional Probability	1	To understand and apply the concept of conditional probability in various dependent event scenarios.	Revise basic probability and read about conditional events	Lecture	Concept explanation with Venn diagrams	Solve 3 real-life based conditional probability questions	31	1	No Linkage	No Linkage	Strong Linkage	No Linkage	No Linkage	No Linkage
32	Probability Distribution	Random variables D	1	To define and distinguish discrete random variables and construct probability mass functions.	Read meaning of discrete random variables	Lecture	Black board explanation + real-world illustrations	Define and list discrete random variables around you	32	1	No Linkage	No Linkage	No Linkage	Strong Linkage	Strong Linkage	Strong Linkage
33	Probability Distribution	Random variables Cont	1	To introduce continuous random variables and probability density functions.	Read about continuous data examples	Lecture	Graph explanation of continuous variables	Distinguish between discrete and continuous examples	33	1	No Linkage	No Linkage	No Linkage	Strong Linkage	Strong Linkage	Strong Linkage
34	Probability Distribution	Examples	1	To solve problems involving discrete and continuous random variables.	Try basic random variable problems	Tutorial	Practice using sample datasets	Complete practice sheet	34	1	No Linkage	No Linkage	No Linkage	Strong Linkage	Strong Linkage	Strong Linkage
35	Probability Distribution	Discrete probability distribution.	1	To understand and compute probabilities using standard discrete distributions.	Watch video tutorial on discrete dist.	Lecture	Explanation using coin and dice examples	Plot a PMF using class data	35	1	No Linkage	No Linkage	No Linkage	Strong Linkage	Strong Linkage	Strong Linkage
36	Probability Distribution	Discrete probability distribution.	1	To reinforce concepts of discrete distributions through applied examples.	Read properties and formulas of discrete distribution	Lecture	Derivations and numerical illustrations	Solve two numerical problems	36	1	No Linkage	No Linkage	No Linkage	Strong Linkage	Strong Linkage	Strong Linkage
37	Probability Distribution	Continuous probability distribution	1	To explain and interpret continuous probability distributions and areas under curves.	Understand PDF and area under curve	Lecture	Graphical method and real-world examples	Draw a curve and calculate basic probabilities	37	1	No Linkage	No Linkage	No Linkage	Strong Linkage	Strong Linkage	Strong Linkage

38	Probability Distribution	Examples	1	To practice problems related to continuous probability functions and their properties.	Try questions on PDF problems	Tutorial	Practice questions from book for mass-exams	Submit solved problems	38	1	No Linkage	No Linkage	No Linkage	Strong Linkage	Strong Linkage	Strong Linkage	
39	Probability Distribution	Binomial and,	1	To understand the structure and applications of the binomial distribution.		Study conditions of binomial distribution	Lecture	Identify binomial experiments in your surroundings		39	1	No Linkage	No Linkage	No Linkage	Strong Linkage	Strong Linkage	Strong Linkage
40	Probability Distribution	Poisson distributions	1	To introduce the Poisson distribution and its use in modeling rare events.		Read definition and assumptions	Lecture	Step-by-step with formula and table usage	Solve 3 problems with different λ values	40	1	No Linkage	No Linkage	No Linkage	Strong Linkage	Strong Linkage	Strong Linkage
41	Probability Distribution	Continuous probability distribution	1	To explore additional applications and examples of continuous probability models.	Review previous PDF concepts	Lecture	Use of standard normal table	Solve area under curve questions	41	1	No Linkage	No Linkage	No Linkage	Strong Linkage	Strong Linkage	Strong Linkage	
42	Probability Distribution	Examples	1	To solve a variety of problems using continuous and discrete probability distributions.	Solve binomial and Poisson examples	Tutorial	Group worksheet solving	Peer assessment of solutions	42	1	No Linkage	No Linkage	No Linkage	Strong Linkage	Strong Linkage	Strong Linkage	
43	Probability Distribution	Normal Distribution	1	To introduce the normal distribution, its properties, and standardization techniques.	Read properties of normal curve	Lecture	Explain using bell curve	Calculate Z scores from class data	43	1	No Linkage	No Linkage	No Linkage	Strong Linkage	Strong Linkage	Strong Linkage	
44	Probability Distribution	Uniform distribution, Exponential distribution	1	To study the uniform and exponential distributions and their applications in real-life scenarios.	Read formulas and graphs of both	Lecture	Visual demonstration and formula derivation	Apply distributions to solve practical problems	44	1	No Linkage	No Linkage	No Linkage	Strong Linkage	Strong Linkage	Strong Linkage	
45	Probability Distribution	Examples	1	To summarize and practice problems across all probability distributions studied.	Solve normal and uniform distribution problems	Tutorial		Write takeaway from each distribution learned	45	1	No Linkage	No Linkage	No Linkage	Strong Linkage	Strong Linkage	Strong Linkage	
							Quiz and revision										